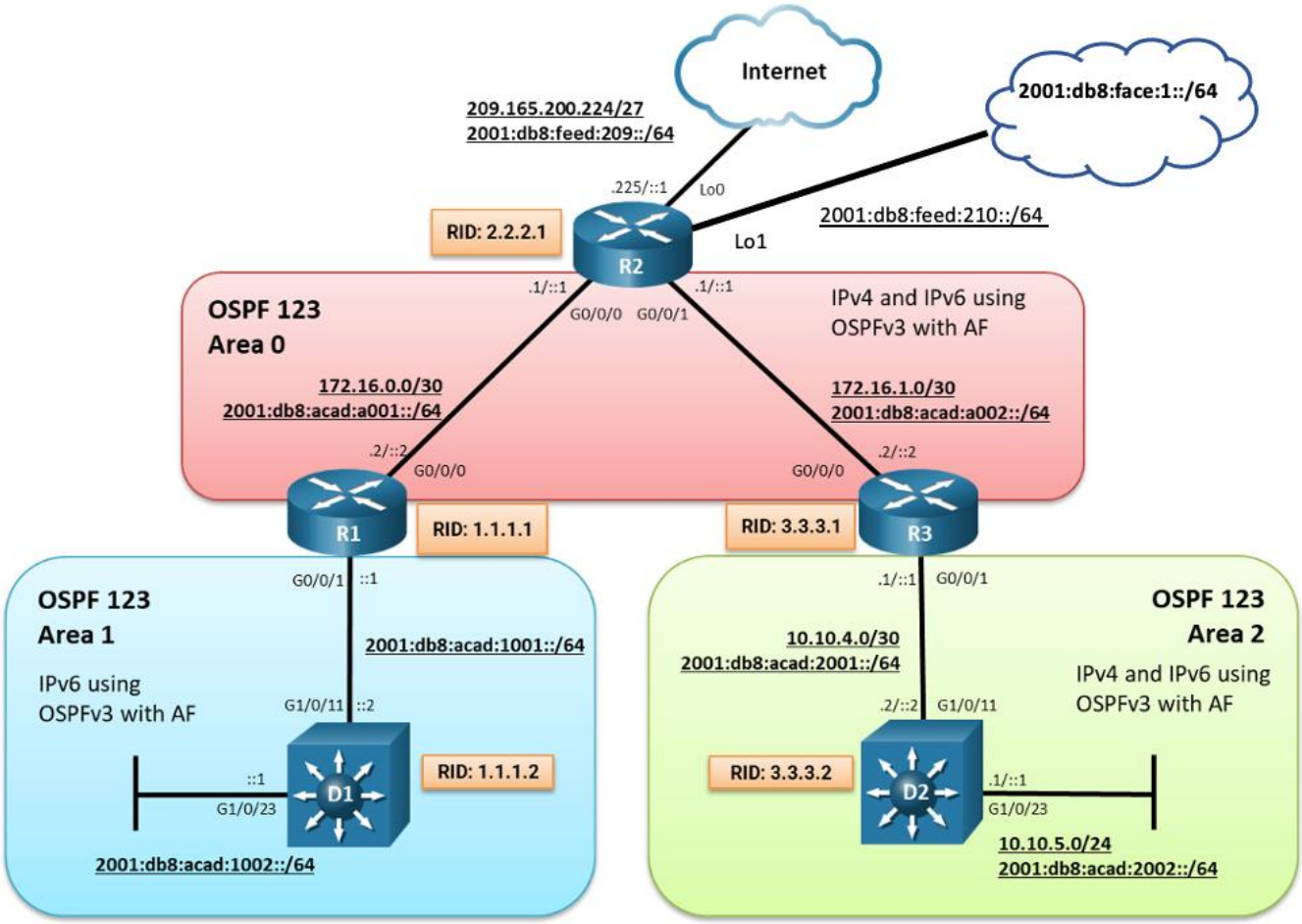
Lab - Configure OSPF Stub Areas and Summarization

Topology



Addressing Table

Device	Interface	IPv4 Address	IPv6 Address	IPv6 Link-Local
R1	G0/0/0	172.16.0.2/30	2001:db8:acad:a001::2/64	fe80::1:2
	G0/0/1	10.10.0.1/30	2001:db8:acad:1001::1/64	fe80::1:1
R2	Lo0	209.165.200.225/27	2001:db8:feed:209::1/64	fe80::2:3
	Lo1	N/A	2001:db8:feed:210::1/64	fe80::2:4
	G0/0/0	172.16.0.1/30	2001:db8:acad:a001::1/64	fe80::2:1
	G0/0/1	172.16.1.1/30	2001:db8:acad:a002::1/64	fe80::2:2
R3	G0/0/0	172.16.1.2/30	2001:db8:acad:a002::2/64	fe80::3:2
	G0/0/1	10.10.4.1/30	2001:db8:acad:2001::1/64	fe80::3:1
D1	G1/0/11	N/A	2001:db8:acad:1001::2/64	fe80::d1:2

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Device	Interface	IPv4 Address	IPv6 Address	IPv6 Link-Local
	G1/0/23	N/A	2001:db8:acad:1002::1/64	fe80::d1:1
D2	G1/0/11	10.10.4.2/30	2001:db8:acad:2001::2/64	fe80::d2:2
	G1/0/23	10.10.5.1/24	2001:db8:acad:2002::1/64	fe80::d2:1

Objectives

Part 1: Build the Topology and Configure Basic Device Settings and IP Addressing

Part 2: Create an IPv6 Static Route on R2 and Redistribute via OSPFv3

Part 3: Understanding OSPFv3 LSAs

Part 4: OSPFv3 Stub Areas

Part 5: OSPFv3 Totally Stubby Areas

Part 6: OSPFv3 NSSA

Part 7: OSPFv3 Totally NSSA

Part 8: Reachability for All Areas

Background / Scenario

In this lab, you will build upon the same **multiarea OSPFv3 topology** introduced in **Lab 7.1.1 - Implement Multiarea OSPFv3**, while focusing on how different OSPF area types control routing information using Link-State Advertisements (LSAs). The physical topology and general roles of the devices remain familiar, allowing you to concentrate on understanding **stub areas, totally stubby areas,** and **Not-So-Stubby Areas (NSSAs)** rather than relearning the network layout.

Several intentional changes have been made compared to Lab 7.1.1:

- **All routers and Layer 3 switches run OSPFv3 using Address Families (AF)** for IPv4 and IPv6, instead of using OSPFv2 for IPv4.
- **All inter-router links are configured as OSPF point-to-point networks**, eliminating Designated Router (DR) and Network LSA behavior and simplifying the analysis of LSAs.
- **No route summarization is configured**, allowing you to clearly observe how individual prefixes are advertised or suppressed as area types change.
- **Router R2 includes an additional IPv6 Loopback interface** and an **IPv6 static route** that is redistributed into OSPFv3, providing a concrete example of external routing information that will behave differently in normal, stub, totally stubby, and NSSA areas.

Throughout this lab, you will examine how OSPFv3 uses LSAs to control routing information distribution while maintaining full end-to-end reachability. By comparing routing tables and LSDB contents across different area types, you will see that OSPF area design affects the **control plane**, not the **data plane**, and that traffic forwarding continues to work even as routing information is selectively filtered.

Note: The routers used with this CCNP hands-on lab are Cisco 8200Ls and the two Layer 3 switches are Catalyst 9300L switches using Cisco IOS XE Release 17.12.05 (universalk9 image). Other routers and Layer 3 switches and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: Ensure that the routers and switches have been erased and have no startup configurations. If you are unsure, contact your instructor.

Required Resources

- 3 Routers (Cisco Catalyst 8200L with Cisco IOS XE Release 17.12.05c universal image or comparable)

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- 2 Switches (Cisco Catalyst 9300L with Cisco IOS XE Release 17.12.05 universal image or comparable)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Instructions

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

In Part 1, you will set up the network topology and configure basic settings and interface addressing on routers and switches.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram and cable as necessary.

Step 2: Configure basic settings for each router.

a. Console into each device, enter global configuration mode, and apply the basic settings and interface addressing using the following startup configurations for each device.

Router R1

```
hostname R1
!
line con 0
exec-timeout 0 0
logging synchronous
no ip domain lookup
!
ipv6 unicast-routing
!
interface GigabitEthernet0/0/0
 ip address 172.16.0.2 255.255.255.252
 ipv6 address FE80::1:2 link-local
 ipv6 address 2001:DB8:ACAD:A001::2/64
 ospfv3 network point-to-point
 ospfv3 123 ipv6 area 0
 ospfv3 123 ipv4 area 0
 no shutdown
 exit
!
interface GigabitEthernet0/0/1
 ipv6 address FE80::1:1 link-local
 ipv6 address 2001:DB8:ACAD:1001::1/64
 ospfv3 network point-to-point
 ospfv3 123 ipv6 area 1
 ospfv3 123 ipv4 area 1
 no shutdown
```

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```
 exit
!
router ospfv3 123
 !
 address-family ipv4 unicast
  router-id 1.1.1.1
 exit-address-family
 !
 address-family ipv6 unicast
  router-id 1.1.1.1
 exit-address-family
!
end
```

Router R2

```
hostname R2
!
line con 0
exec-timeout 0 0
logging synchronous
no ip domain lookup
!
ipv6 unicast-routing
!
interface Loopback0
 ip address 209.165.200.225 255.255.255.224
 ipv6 address FE80::2:3 link-local
 ipv6 address 2001:DB8:FEED:209::1/64
 no shutdown
 exit
!
interface Loopback1
 ipv6 address FE80::2:4 link-local
 ipv6 address 2001:DB8:FEED:210::1/64
 no shutdown
 exit
!
interface GigabitEthernet0/0/0
 ip address 172.16.0.1 255.255.255.252
 ipv6 address FE80::2:1 link-local
 ipv6 address 2001:DB8:ACAD:A001::1/64
 ospfv3 network point-to-point
 ospfv3 123 ipv6 area 0
 ospfv3 123 ipv4 area 0
 no shutdown
 exit
```

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```
!
interface GigabitEthernet0/0/1
 ip address 172.16.1.1 255.255.255.252
 ipv6 address FE80::2:2 link-local
 ipv6 address 2001:DB8:ACAD:A002::1/64
 ospfv3 network point-to-point
 ospfv3 123 ipv6 area 0
 ospfv3 123 ipv4 area 0
 no shutdown
 exit
!
router ospfv3 123
 !
 address-family ipv4 unicast
  default-information originate
  router-id 2.2.2.1
 exit-address-family
 !
 address-family ipv6 unicast
  default-information originate
  router-id 2.2.2.1
 exit-address-family
!
ip route 0.0.0.0 0.0.0.0 Loopback0
!
ipv6 route ::/0 Loopback0
!
end
```

Router R3

```
hostname R3
!
line con 0
exec-timeout 0 0
logging synchronous
no ip domain lookup
!
ipv6 unicast-routing
!
interface GigabitEthernet0/0/0
 ip address 172.16.1.2 255.255.255.252
 ipv6 address FE80::3:2 link-local
 ipv6 address 2001:DB8:ACAD:A002::2/64
 ospfv3 network point-to-point
 ospfv3 123 ipv6 area 0
 ospfv3 123 ipv4 area 0
```

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```
 no shutdown
 exit
!
interface GigabitEthernet0/0/1
 ip address 10.10.4.1 255.255.255.252
 ipv6 address FE80::3:1 link-local
 ipv6 address 2001:DB8:ACAD:2001::1/64
 ospfv3 network point-to-point
 ospfv3 123 ipv6 area 2
 ospfv3 123 ipv4 area 2
 no shutdown
 exit
!
router ospfv3 123
!
 address-family ipv4 unicast
 router-id 3.3.3.1
 exit-address-family
 !
 address-family ipv6 unicast
 router-id 3.3.3.1
 exit-address-family
!
end
```

Switch D1

```
hostname D1
!
line con 0
exec-timeout 0 0
logging synchronous
no ip domain lookup
!
ip routing
!
ipv6 unicast-routing
!
interface GigabitEthernet1/0/11
 no switchport
 ipv6 address FE80::D1:2 link-local
 ipv6 address 2001:DB8:ACAD:1001::2/64
 ospfv3 network point-to-point
 ospfv3 123 ipv6 area 1
 no shutdown
 exit
```

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```
!
interface GigabitEthernet1/0/23
 no switchport
 ipv6 address FE80::D1:1 link-local
 ipv6 address 2001:DB8:ACAD:1002::1/64
 ospfv3 123 ipv6 area 1
 no shutdown
 exit
!
router ospfv3 123
!
 address-family ipv6 unicast
  passive-interface GigabitEthernet1/0/23
  router-id 1.1.1.2
 exit-address-family
!
end
```

Switch D2

```
hostname D2
!
line con 0
exec-timeout 0 0
logging synchronous
no ip domain lookup
!
ip routing
!
ipv6 unicast-routing
!
interface GigabitEthernet1/0/11
 no switchport
 ip address 10.10.4.2 255.255.255.252
 ipv6 address FE80::D2:2 link-local
 ipv6 address 2001:DB8:ACAD:2001::2/64
 ospfv3 network point-to-point
 ospfv3 123 ipv6 area 2
 ospfv3 123 ipv4 area 2
 no shutdown
 exit
!
interface GigabitEthernet1/0/23
 no switchport
 ip address 10.10.5.1 255.255.255.0
 ipv6 address FE80::D2:1 link-local
```

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```
 ipv6 address 2001:DB8:ACAD:2002::1/64
 ospfv3 123 ipv6 area 2
 ospfv3 123 ipv4 area 2
 no shutdown
 exit
!
router ospfv3 123
 !
 address-family ipv4 unicast
  passive-interface GigabitEthernet1/0/23
  router-id 3.3.3.2
 exit-address-family
 !
 address-family ipv6 unicast
  passive-interface GigabitEthernet1/0/23
  router-id 3.3.3.2
 exit-address-family
!
end
```

b. Save the running configuration to startup-config.

Part 2: Create an IPv6 Static Route on R2 and Redistribute via OSPFv3

In this part, you will configure an IPv6 static route on router R2 and redistribute it into OSPFv3. This redistributed route will be advertised as an **OSPF external route**, allowing you to observe how OSPFv3 handles routes that originate outside the OSPF domain. Creating this external route now provides a concrete example that will be used later in the lab when discussing **LSA Type 5** behavior and how external routes are treated differently in normal, stub, totally stubby, and NSSA areas.

Step 1: Configure an IPv6 Static route.

a. On the ASBR router R2, configure an IPv6 static route on R2 to 2001:db8:face:1::/64 using the exit interface Loopback 1. This static route represents an external network reachable through R2.

```
R2(config)# ipv6 route 2001:db8:face:1::/64 loopback1
```

b. Verify static route on R2.

```
R2# show ipv6 route static | begin ::/0
S   ::/0 [1/0]
     via Loopback0, directly connected
S   2001:DB8:FACE:1::/64 [1/0]
     via Loopback1, directly connected
```

Step 2: Redistribute the IPv6 Static route via OSPFv3.

On R2, redistribute the IPv6 static route using OSPFv3 with AF.

```
R2(config)# router ospfv3 123
R2(config-router)# address-family ipv6 unicast
R2(config-router-af)# redistribute ?
```

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```
  application     Application Protocol
  bgp             Border Gateway Protocol (BGP)
  connected       Connected Routes
  eigrp           Enhanced Interior Gateway Routing Protocol (EIGRP)
  isis            ISO IS-IS
  lisp            Locator ID Separation Protocol (LISP)
  maximum-prefix  Maximum number of prefixes redistributed to protocol
  nat-route       NAT-Route
  nd              Neighbor Discovery
  omp             Overlay Management Protocol (OMP)
  ospf            Open Shortest Path First (OSPFv3)
  rip             IPv6 Routing Information Protocol (RIPv6)
  static          Static Routes

R2(config-router-af)# redistribute static
R2(config-router-af)# end
```

Step 3: Verify the IPv6 Static Route was Redistributed using OSPFv3

All of the redistributed IPv6 routes appear as **OE2 routes with a cost of 20** because OSPF assigns a **default external metric of 20** to routes redistributed into OSPF when no metric is explicitly configured. For **Type 2 external routes (E2)**, this external metric remains the same throughout the OSPF domain and does not increase as the route is advertised to other routers.

Even though all routers see the same external cost, OSPF still uses its internal topology information, carried in **Router LSAs (Type 1)** and **Inter-Area Prefix LSAs (Type 3)**, to determine the best path within the OSPF domain to reach the ASBR.

With OSPF external Type 2 routes, multiple ABRs can advertise the same external prefix with the same cost. The selected path may not reflect the lowest true end-to-end cost, because it only considers the cost to reach the advertising router. In network designs with multiple exit points and a need for accurate path selection, Type 1 external routes are often preferred because they include the internal OSPF cost in the metric calculation.

a. Verify the redistributed static route was received on the ABR router, R1.

```
R1# show ipv6 route ospf | begin OE2
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       a - Application, m - OMP
OE2 ::/0 [110/1], tag 123
     via FE80::2:1, GigabitEthernet0/0/0
O   2001:DB8:ACAD:1002::/64 [110/2]
     via FE80::D1:2, GigabitEthernet0/0/1
OI  2001:DB8:ACAD:2001::/64 [110/3]
     via FE80::2:1, GigabitEthernet0/0/0
OI  2001:DB8:ACAD:2002::/64 [110/4]
     via FE80::2:1, GigabitEthernet0/0/0
O   2001:DB8:ACAD:A002::/64 [110/2]
     via FE80::2:1, GigabitEthernet0/0/0
OE2 2001:DB8:FACE:1::/64 [110/20]
     via FE80::2:1, GigabitEthernet0/0/0
```

b. Verify the redistributed static route was received on the ABR router, R3.

```
R3# show ipv6 route ospf | begin OE2
```

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```
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       a - Application, m - OMP
OE2 ::/0 [110/1], tag 123
     via FE80::2:2, GigabitEthernet0/0/0
OI  2001:DB8:ACAD:1001::/64 [110/3]
     via FE80::2:2, GigabitEthernet0/0/0
OI  2001:DB8:ACAD:1002::/64 [110/4]
     via FE80::2:2, GigabitEthernet0/0/0
O   2001:DB8:ACAD:2002::/64 [110/2]
     via FE80::D2:2, GigabitEthernet0/0/1
O   2001:DB8:ACAD:A001::/64 [110/2]
     via FE80::2:2, GigabitEthernet0/0/0
OE2 2001:DB8:FACE:1::/64 [110/20]
     via FE80::2:2, GigabitEthernet0/0/0
```

c. Verify the redistributed static route was received on the internal Area 1 router, D1.

```
D1# show ipv6 route ospf | begin OE2
       OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1
       ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-registrations
       ld - LISP dyn-eid, lA - LISP away, le - LISP extranet-policy
       lp - LISP publications, ls - LISP destinations-summary
OE2 ::/0 [110/1], tag 123
     via FE80::1:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:2001::/64 [110/4]
     via FE80::1:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:2002::/64 [110/5]
     via FE80::1:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:A001::/64 [110/2]
     via FE80::1:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:A002::/64 [110/3]
     via FE80::1:1, GigabitEthernet1/0/11
OE2 2001:DB8:FACE:1::/64 [110/20]
     via FE80::1:1, GigabitEthernet1/0/11
```

d. Verify the redistributed static route was received on the internal Area 2 router, D2.

```
D2# show ipv6 route ospf | begin OE2
       OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1
       ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-registrations
       ld - LISP dyn-eid, lA - LISP away, le - LISP extranet-policy
       lp - LISP publications, ls - LISP destinations-summary
OE2 ::/0 [110/1], tag 123
     via FE80::3:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:1001::/64 [110/4]
     via FE80::3:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:1002::/64 [110/5]
     via FE80::3:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:A001::/64 [110/3]
     via FE80::3:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:A002::/64 [110/2]
     via FE80::3:1, GigabitEthernet1/0/11
```

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```
OE2 2001:DB8:FACE:1::/64 [110/20]
     via FE80::3:1, GigabitEthernet1/0/11
```

Step 4: Verify the IPv4 Routing Tables

All OSPF routers, except D1 which is in an IPv6-only area, should have an IPv4 default route redistributed via OSPF, along with knowledge of intra-area OSPF (O) and inter-area OSPF (OI) routes. On the ABRs and internal OSPF routers, notice any internal OSPF routes (O), OSPF inter-area (O IA) routes from other areas and the IPv4 route redistributed (External Type 2) via OSPFv3.

a. Verify the IPv4 routing table on the ASBR, R2. R2 should contain an IPv4 static default route and all inter-area OSPF routes (O IA).

```
R2# show ip route | begin Gateway
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

S*    0.0.0.0/0 is directly connected, Loopback0
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
O IA     10.10.4.0/30 [110/2] via 172.16.1.2, 02:41:44, GigabitEthernet0/0/1
O IA     10.10.5.0/24 [110/3] via 172.16.1.2, 02:21:36, GigabitEthernet0/0/1
      172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C        172.16.0.0/30 is directly connected, GigabitEthernet0/0/0
L        172.16.0.1/32 is directly connected, GigabitEthernet0/0/0
C        172.16.1.0/30 is directly connected, GigabitEthernet0/0/1
L        172.16.1.1/32 is directly connected, GigabitEthernet0/0/1
      209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C        209.165.200.224/27 is directly connected, Loopback0
L        209.165.200.225/32 is directly connected, Loopback0
```

b. Verify OSPF routes in the IPv4 routing table on the ABR, R1.

```
R1# show ip route ospfv3 | begin Gateway
Gateway of last resort is 172.16.0.1 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 172.16.0.1, 03:23:29, GigabitEthernet0/0/0
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
O IA     10.10.4.0/30 [110/3] via 172.16.0.1, 03:22:36, GigabitEthernet0/0/0
O IA     10.10.5.0/24 [110/4] via 172.16.0.1, 03:02:27, GigabitEthernet0/0/0
      172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O        172.16.1.0/30 [110/2] via 172.16.0.1, 03:23:29, GigabitEthernet0/0/0
```

c. Verify OSPF routes in the IPv4 routing table on the ABR, R3.

```
R3# show ip route ospfv3 | begin Gateway
Gateway of last resort is 172.16.1.1 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 172.16.1.1, 02:40:33, GigabitEthernet0/0/0
      10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
O        10.10.5.0/24 [110/2] via 10.10.4.2, 02:20:24, GigabitEthernet0/0/1
      172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O        172.16.0.0/30 [110/2] via 172.16.1.1, 02:40:33, GigabitEthernet0/0/0
```

d. Verify that there are no OSPF routes in the IPv4 routing table on the Area 1, D1. This is because this area is IPv6 only.

```
D1# show ip route ospfv3 | begin Gateway
```

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```
D1#
```

e. Verify OSPF routes in the IPv4 routing table on the Area 2, D2.

```
D2# show ip route ospfv3 | begin Gateway
Gateway of last resort is 10.10.4.1 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.10.4.1, 02:20:45, GigabitEthernet1/0/11
      172.16.0.0/30 is subnetted, 2 subnets
O IA     172.16.0.0 [110/3] via 10.10.4.1, 02:20:45, GigabitEthernet1/0/11
O IA     172.16.1.0 [110/2] via 10.10.4.1, 02:20:45, GigabitEthernet1/0/11
```

Part 3: Understanding OSPFv3 LSAs

Understanding OSPFv3 LSAs is essential to understanding how different OSPF area types behave. Normal, stub, totally stubby, NSSA, and totally NSSA areas are not defined by different routing protocols, but by **which LSA types are allowed into an area and how they are handled by ABRs**.

By examining the purpose of each LSA type, you can predict which routes will appear in the routing table, which routes will be suppressed, and why default routes are injected into certain area types. This part of the lab provides the foundation you need to understand how OSPFv3 uses LSAs to control routing information as area types change.

The following table summarizes the most common OSPF LSA types and shows how their names and roles differ between OSPFv2 and OSPFv3. While many LSAs serve the same fundamental purpose in both versions, OSPFv3 uses updated names that better reflect their function, particularly in a multi-area and address-family environment.

The table includes an **NSSA External LSA (Type 7)** which will be discussed later when we configure a Not-So-Stubby-Area. The table also includes two additional OSPFv3-specific LSAs, **Link LSA (Type 8)** and **Intra-Area Prefix LSA (Type 9)**, which were introduced to support IPv6 and address families.

LSA Type	OSPFv2 Name	OSPFv3 Name	Generated By	Purpose / Description
1	Router LSA	Router LSA	**All OSPF routers**	Describes router links and the topology within an area. In OSPFv3, it carries **topology only** (no prefixes).
2	Network LSA	Network LSA	**Designated Router (DR)** on multiaccess networks	Lists all routers on a multiaccess segment. No prefixes in OSPFv3.
3	Network Summary LSA	Inter-Area Prefix LSA	**ABRs**	Advertises **inter-area prefixes** into other areas. It carries prefix information only.
4	ASBR Summary LSA	Inter-Area Router LSA	**ABRs**	Advertises reachability to an **ASBR** so other areas can locate it.
5	AS-External LSA	AS-External LSA	**ASBRs**	Advertises routes redistributed into OSPF. Flooded throughout the domain (not allowed in stub areas).
7	NSSA External LSA	Type-7 LSA	**ASBRs inside an NSSA**	Advertises external routes inside an NSSA; translated to Type 5 by the ABR.
8	—	**Link LSA (Type 8)**	**All OSPFv3 routers (per link)**	Advertises link-local information (IPv6 addresses). Flooded **only on the local link**.
9	—	**Intra-Area Prefix LSA (Type 9)**	**All OSPFv3 routers (and DRs for networks)**	Carries IPv4 and/or IPv6 prefixes associated with routers or networks within an area.

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Note: In OSPFv3, **topology LSAs (Types 1 and 2)** and **prefix LSAs (Type 9)** are generated separately, allowing a single LSDB to support multiple address families.

Types of LSAs

Below is more detailed information for each type of LSA:

LSA Type 1

OSPFv2: Router LSA

OSPFv3: Router LSA (0x2001)

The Router LSA describes a router's links and neighbors within an area and forms the foundation of the OSPF topology. In OSPFv3, this LSA carries **topology information only**, with all prefix information separated into other LSAs.

Note: In **OSPFv2**, the Router LSA (Type 1) carries both **topology information** and the **IPv4 prefixes** associated with the router's interfaces. In **OSPFv3**, the Router LSA carries **topology information only**. All IPv4 and IPv6 prefixes within the area are advertised separately using Intra-Area Prefix LSAs (Type 9).

Summary:

- Describes the router's links and neighbors within an area
- Flooded within the area only
- No prefixes in OSPFv3 (topology only)

LSA Type 2

OSPFv2: Network LSA

OSPFv3: Network LSA (0x2002)

The Network LSA is generated by the Designated Router (DR) on multiaccess networks and lists all routers connected to that segment. In OSPFv2, this LSA also carried the IPv4 network prefix, but in OSPFv3 it carries topology information only, with all prefixes advertised separately using Intra-Area Prefix LSAs (Type 9).

Summary:

- Generated by the DR on multiaccess networks
- Lists all routers attached to that segment
- Flooded within the area only
- No prefixes in OSPFv3

LSA Type 3

OSPFv2: Network Summary LSA

OSPFv3: Inter-Area Prefix LSA (0x2003)

The Inter-Area Prefix LSA is generated by Area Border Routers (ABRs) to advertise networks from one area into another. In OSPFv3, this LSA carries **only prefix information**,.

Summary:

- Generated by ABRs
- Advertises inter-area prefixes
- Flooded into other areas

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- Carries only prefix information in OSPFv3

LSA Type 4

OSPFv2: ASBR Summary LSA

OSPFv3: Inter-Area Router LSA (0x2004)

The Inter-Area Router LSA is generated by ABRs to advertise how to reach an Autonomous System Boundary Router (ASBR). This LSA allows routers in other areas to locate the ASBR that is injecting external routes.

Summary:

- Generated by ABRs
- Advertises reachability to an ASBR
- Used when external routes exist
- Carries router reachability, not prefixes

LSA Type 5

OSPFv2: AS-External LSA

OSPFv3: AS-External LSA (0x2005)

The AS-External LSA advertises routes that have been redistributed into OSPF from outside the OSPF domain. These LSAs are flooded throughout the OSPF domain but are **not permitted in stub or totally stubby areas**.

Summary:

- Generated by ASBRs
- Advertises routes redistributed into OSPF
- Flooded throughout the OSPF domain
- Blocked by stub and totally stubby areas

LSA Type 7

OSPFv2: NSSA External LSA

OSPFv3: Type-7 LSA (0x2007)

The Type-7 LSA is used within Not-So-Stubby Areas (NSSAs) to carry external routes without flooding Type 5 LSAs into the area. These LSAs are translated to Type 5 by the ABR when advertised into other areas, unless translation is explicitly disabled.

Summary:

- Used inside NSSA areas
- Allows externals without flooding Type 5 LSAs
- Translated to Type 5 by the ABR (unless suppressed)

How LSAs are Used to Identify an ASBR and Advertises External Routes

When a router is configured to **originate a default route** or **redistribute routes** into OSPF, it becomes an **Autonomous System Boundary Router (ASBR)**. As part of this process, the router sets the **external bit (E-bit)** in its **Router LSA (Type 1)** to indicate that it is injecting external routes into the OSPF domain.

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ABRs receive this Router LSA and recognize the router as an ASBR based on the set E-bit. The ABRs then advertise reachability to that ASBR by generating **Inter-Area Router LSAs (Type 4)** and flooding them into other areas. These LSA Type 4s allow routers in different areas to know how to reach the ASBR itself.

The actual external routes, such as the redistributed default route and the redistributed IPv6 static route, are carried in **AS-External LSAs (Type 5)**, which are flooded throughout the OSPF routing domain (except into stub areas). Together, LSA Type 4 and LSA Type 5 allow routers to both **locate the ASBR** and **learn the external destinations** it advertises.

OSPFv3-Specific LSAs 8 and 9: Link and Intra-Area Prefix Information

OSPFv3 introduced two additional LSAs: **LSA Type 8 (Link LSA)** and **LSA Type 9 (Intra-Area Prefix LSA)**. These LSAs were added support IPv6 and the use of multiple address families within a single OSPF process. Unlike earlier LSAs that describe network topology and external reachability, these LSAs focus on **local link information** and **prefix information** that was previously embedded directly in topology LSAs. Understanding LSA Types 8 and 9 is vital to understanding how OSPFv3 separates topology from addressing while maintaining the same fundamental SPF behavior as OSPFv2.

OSPFv3 LSA Type 8: Link LSA

The introduction of the Link LSA fulfills a requirement unique to IPv6 that did not exist in the IPv4-centric world of OSPFv2.

Why is there no LSA Type 8 in OSPFv2

OSPFv2 was designed exclusively for IPv4 networks. In IPv4, interfaces are not required to use link-local addresses, and routers can form neighbor relationships using globally-significant IPv4 addresses. Because of this, there was no need for OSPFv2 to explicitly advertise per-link addressing information. All required information for neighbor formation and routing could be inferred from the existing LSAs.

What changed with OSPFv3

OSPFv3 was designed to support IPv6. In IPv6, every interface is required to have a link-local address, and OSPFv3 uses these link-local addresses to form adjacencies and exchange routing information. As a result, routers on the same link must know each other's IPv6 link-local addresses to communicate correctly.

The solution: LSA Type 8 (Link LSA)

To support this requirement, OSPFv3 introduced **LSA Type 8**, known as the Link LSA. This LSA allows a router to advertise its IPv6 link-local address and other link-specific information to neighbors on the same link. The Link LSA is flooded only on the local link and is never propagated beyond that link.

Why this matters

LSA Type 8 enables OSPFv3 to function correctly over IPv6 by supporting neighbor discovery and adjacency formation using link-local addresses. By keeping this information local to the link, OSPFv3 avoids unnecessary flooding and maintains efficient, scalable operation.

OSPFv3 LSA Type 9: Intra-Area Prefix LSA

To transition from a single-protocol design to a multi-address-family environment, OSPFv3 required a new mechanism to carry prefix information independent of topology information.

Why no LSA Type 9 in OSPFv2

In OSPFv2, topology information and prefix information were combined within the same LSAs. Router LSAs and Network LSAs described both how routers were connected and which IPv4 networks were reachable. This design tightly coupled OSPFv2 to IPv4 and made the protocol less flexible. Supporting additional address families would have required significant architectural changes.

What changed with OSPFv3

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OSPFv3 was designed to support IPv6 from the start and was later extended to carry IPv4 routes using address families. A single OSPFv3 process must be able to support multiple address families simultaneously. Continuing to combine topology and prefix information would have limited scalability and flexibility in this multi-address-family environment.

The solution: LSA Type 9 (Intra-Area Prefix LSA)

To address this, OSPFv3 introduced **LSA Type 9**, known as the Intra-Area Prefix LSA. In OSPFv3, topology information is carried by Router LSAs (Type 1), while all IPv4 and IPv6 prefixes within an area are carried separately in Intra-Area Prefix LSAs (Type 9).

This LSA carries prefix information only and is used to advertise IPv4 and IPv6 prefixes within an area. Prefixes advertised in a Type 9 LSA are associated with either a Router LSA or a Network LSA, while the topology itself remains unchanged.

LSA Type 9 is flooded only within the local area and is used by all address families supported by OSPFv3.

Why this matters

Separating prefix information from topology information allows OSPFv3 to support multiple address families cleanly and efficiently. SPF calculations are performed using the topology described in LSA Type 1 (and Type 2 when present), and prefixes from LSA Type 9 are applied to the resulting shortest-path tree. This design makes OSPFv3 more flexible, scalable, and future proof than OSPFv2.

Normal OSPF Areas

Before we discuss stub and other area types, we need to review normal OSPF areas.

A **normal OSPF area** is the default OSPF area type and allows the full exchange of routing information. Routers in a normal area can receive **intra-area routes, inter-area routes, and external routes**, since no LSA types are filtered by the ABR.

D2 is an internal router in a normal OSPF area. Examining D2's IPv4 and IPv6 routing tables, D2 is receiving all OSPF IPv4 and IP6 routes from all areas, an IPv4 and IPv6 default route from the ASBR, and a static IPv4 route redistributed from the ASBR. No routes are being blocked by R1, the ABR.

```
D2# show ip route ospfv3 | begin Gateway
Gateway of last resort is 10.10.4.1 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.10.4.1, 10:22:30, GigabitEthernet1/0/11
      172.16.0.0/30 is subnetted, 2 subnets
O IA     172.16.0.0 [110/3] via 10.10.4.1, 10:22:30, GigabitEthernet1/0/11
O IA     172.16.1.0 [110/2] via 10.10.4.1, 10:22:30, GigabitEthernet1/0/11
D2# show ipv6 route ospf | begin OE2
       OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1
       ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-registrations
       ld - LISP dyn-eid, lA - LISP away, le - LISP extranet-policy
       lp - LISP publications, ls - LISP destinations-summary
OE2 ::/0 [110/1], tag 123
     via FE80::3:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:1001::/64 [110/4]
     via FE80::3:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:1002::/64 [110/5]
     via FE80::3:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:A001::/64 [110/3]
     via FE80::3:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:A002::/64 [110/2]
```

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```
     via FE80::3:1, GigabitEthernet1/0/11
OE2 2001:DB8:FACE:1::/64 [110/20]
     via FE80::3:1, GigabitEthernet1/0/11
```

Which LSAs Is D2 Receiving (and What Each Explains)

In a normal OSPFv3 area, there is no blocking of LSAs. D2 receives **Type 1 LSAs** that describe intra-area topology, **Type 9 LSAs** that carry intra-area IPv4 and IPv6 prefixes, **Type 3 LSAs** for inter-area prefixes, and because an ASBR exists, **Type 4 LSAs** to locate the ASBR and **Type 5 LSAs** that advertise external routes such as the default and redistributed static route.

D2's OSPFv3 LSAs

- **LSA Type 1 (Router LSA)**

  D2 receives Type 1 LSAs for Area 2, which describe the **topology inside the area**, including which routers are connected and the cost of each link. These LSAs are used by D2 to run the SPF algorithm and build the **shortest-path tree** toward other routers in the same area, including its ABR (R3).

- **LSA Type 9 (Intra-Area Prefix LSA)**

  Because this is OSPFv3, D2 also receives Type 9 LSAs, which carry the **IPv4 and IPv6 prefixes associated with the intra-area topology**. These prefixes are linked to the Router LSAs (and Network LSAs, if present), allowing D2 to populate its routing table after the SPF calculation is complete.

- **LSA Type 3 (Inter-Area Prefix LSA)**

  D2 receives Type 3 LSAs from the ABR (R3), which advertise prefixes that originate in **other areas**, such as Area 0 and Area 1. These LSAs explain why D2 installs **inter-area routes (O IA / OI)** in both its IPv4 and IPv6 routing tables.

- **LSA Type 4 (Inter-Area Router LSA)**

  Because R2 acts as an **ASBR** by injecting external routes, D2 must know how to reach that ASBR. The ABR, R3, injects Type 4 LSAs into Area 2 to provide reachability information to the **ASBR's router ID**. This enables internal routers such as D2 to determine the correct internal path to the ASBR R2.

- **LSA Type 5 (AS-External LSA)**

  D2 receives Type 5 LSAs from the ASBR R2 and forwarded by the ABR R3 that advertise routes redistributed into OSPF, including:

  - The redistributed **default route** (IPv4 0.0.0.0/0 and IPv6 ::/0)
  - The redistributed IPv6 static route 2001:db8:face:1::/64

  These LSAs explain why D2 sees **O*E2 / OE2 routes** when operating in a normal area.

- **LSA Type 8 (Link LSA)**

  D2 also exchanges Type 8 LSAs on each OSPFv3-enabled link. These LSAs advertise **IPv6 link-local addressing information** used for neighbor communication. While Type 8 LSAs do not influence route selection, they are essential for **OSPFv3 adjacency formation and operation**.

Note: To better understand how OSPF operates, the **show ospfv3 database** command can be used to display a summary of the **Link-State Database (LSDB)** and the LSAs it contains. In OSPFv3, a **single LSDB is maintained per area**, and this LSDB is shared by both the IPv4 and IPv6 address families. While routes are installed into separate IPv4 and IPv6 routing tables, the underlying topology and LSA information are learned and stored in a common OSPFv3 database.

Part 4: OSPFv3 Stub Areas

In this part, we will configure Area 2 as an OSPFv3 stub area.

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A **stub area** is an OSPF area designed to **limit external routing information** to reduce routing table size and LSA flooding. Instead of allowing external routes into the area, the Area Border Router (ABR) provides a **default route**, allowing routers inside the stub area to reach destinations outside the area without needing full external route knowledge. In OSPFv3, this behavior is controlled by **which LSA types are permitted or blocked** within the area.

LSAs Allowed and Blocked in an OSPFv3 Stub Area

In an OSPFv3 stub area, routers continue to receive LSAs required to understand local topology and reach other OSPF areas, but **external LSAs are suppressed**.

- **Intra-area LSAs Allowed**
    - **LSA Type 1 (Router LSA)**: Describes intra-area topology
    - **LSA Type 8 (Link LSA)**: Supports IPv6 neighbor communication
    - **LSA Type 9 (Intra-Area Prefix LSA)**: Carries IPv4 and IPv6 prefixes within the area
- **Inter-area LSAs Allowed**
    - **LSA Type 3 (Inter-Area Prefix LSA)**: Advertised by the ABR for networks in other OSPF areas
    - **Default Summary LSA (Type 3)**: Injected by the ABR to provide a default route
- **LSAs Blocked in an OSPFv3 Stub Area**
    - **LSA Type 4 (Inter-Area Router LSAs)**: Because external routes are not allowed into a stub area, there is no need to advertise how to reach an ASBR.
    - **LSA Type 5 (AS-External LSA)**: External routes are blocked in stub areas.

What We Expect to See in a Stub Area

When Area 2 is configured as a stub area, routers inside the area (such as D2) should have:

- Routes to **OSPF networks within Area 2**
- Routes to **OSPF networks in other areas** (learned via Type 3 LSAs)
- A **default route injected by the ABR**
    - This default route allows traffic to exit the stub area
    - It is **not** the default route originally advertised by the ASBR

What We Do Not Expect to See in a Stub Area

Because external LSAs are blocked:

- The **default route advertised by the ASBR** should not appear
- **External routes**, such as the IPv6 static route redistributed by the ASBR and the IPv4/IPv6 default static routes advertised by the ASBR, should not be present.

Summary

In a stub area, OSPF blocks external routes and relies on the ABR-injected default route, allowing routers to reach external destinations without carrying full external routing information.

Step 1: Configure Area 2 as a stub area on ABR router R3.

Both the ABR and all internal routers are configured as part of a stub area by applying the **area** *<area-id>* **stub** command under the appropriate **OSPFv3 address family**, ensuring consistent stub area behavior

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across the area. As you configure this on R3, you will receive messages that neighbor adjacency is forced to reset with D2.

```
R3(config)# router ospfv3 123
R3(config-router)# address-family ipv4 unicast
R3(config-router-af)# area 2 stub
*Jan 16 14:45:11.910: %OSPFv3-5-ADJCHG: Process 123, IPv4, Nbr 3.3.3.2 on
GigabitEthernet0/0/1 from FULL to DOWN, Neighbor Down: Adjacency forced to reset
R3(config-router-af)# exit-address-family
R3(config-router)# address-family ipv6 unicast
R3(config-router-af)# area 2 stub
*Jan 16 14:45:25.943: %OSPFv3-5-ADJCHG: Process 123, IPv6, Nbr 3.3.3.2 on
GigabitEthernet0/0/1 from FULL to DOWN, Neighbor Down: Adjacency forced to reset
R3(config-router-af)# exit-address-family
```

Step 2: Configure Area 2 as a stub area on internal router D2.

Repeat this command on the internal OSPF router, D2. After configuring each address family, you should receive messages that adjacency is reestablished with R3.

```
D2(config)# router ospfv3 123
D2(config-router)# address-family ipv4 unicast
D2(config-router-af)# area 2 stub
*Jan 16 14:50:57.397: %OSPFv3-5-ADJCHG: Process 123, IPv4, Nbr 3.3.3.1 on
GigabitEthernet1/0/11 from LOADING to FULL, Loading Done
D2(config-router-af)# exit-address-family
D2(config-router)# address-family ipv6 unicast
D2(config-router-af)# area 2 stub
*Jan 16 14:51:13.535: %OSPFv3-5-ADJCHG: Process 123, IPv6, Nbr 3.3.3.1 on
GigabitEthernet1/0/11 from LOADING to FULL, Loading Done
D2(config-router-af)# exit-address-family
```

Step 3: Verify routing tables on D2.

After Area 2 is configured as a stub area, D2 continues to receive **intra-area OSPFv3 information**, including topology LSAs (**Type 1**), link-local neighbor information (**Type 8**), and intra-area prefix information (**Type 9**). D2 also receives **inter-area routes** advertised by the ABR R3 through **Type 3 LSAs**, which explains the presence of OSPF inter-area routes in both the IPv4 and IPv6 routing tables.

As expected in a stub area, **external routes advertised via Type 5 LSAs are no longer present**, including the default route and IPv6 static route originally injected by the ASBR. Instead, the **default route now seen on D2 is an inter-area route injected by the ABR (R3) using a Type 3 Summary LSA**, allowing traffic to exit the stub area without carrying external routing information.

Note: Intra-area OSPFv3 routes **(O)** would be included, however, there are only directly connected networks for D2 in Area 2.

```
D2# show ip route ospfv3 | begin Gateway
Gateway of last resort is 10.10.4.1 to network 0.0.0.0

O*IA  0.0.0.0/0 [110/2] via 10.10.4.1, 00:09:13, GigabitEthernet1/0/11
      172.16.0.0/30 is subnetted, 2 subnets
O IA     172.16.0.0 [110/3] via 10.10.4.1, 00:09:13, GigabitEthernet1/0/11
O IA     172.16.1.0 [110/2] via 10.10.4.1, 00:09:13, GigabitEthernet1/0/11
D2# show ipv6 route ospf | begin ::/0
```

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```
OI  ::/0 [110/2]
     via FE80::3:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:1001::/64 [110/4]
     via FE80::3:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:1002::/64 [110/5]
     via FE80::3:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:A001::/64 [110/3]
     via FE80::3:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:A002::/64 [110/2]
     via FE80::3:1, GigabitEthernet1/0/11
```

Although Area 2 is now configured as a stub area, **overall network reachability has not changed**. Internal routers in the stub area use the **default route injected by the ABR** to reach destinations outside the area, while the ABR itself still receives the **external routes and default route advertised by the ASBR**. As a result, traffic can still reach all external networks, even though the detailed external routing information is no longer carried inside the stub area.

This demonstrates that changes to the **control plane** through LSA filtering do not impact the **data plane** and end-to-end traffic forwarding remains unchanged.

Part 5: OSPFv3 Totally Stubby Areas

In this part, you will configure **Area 2 as an OSPFv3 totally stubby area** and examine how this further restricts routing information compared to a standard stub area. A totally stubby area is designed to minimize routing table size by blocking not only external routes, but also inter-area routes, while still preserving full network reachability using a default route injected by the ABR.

In OSPFv3, totally stubby area behavior is achieved by controlling which LSA types are allowed into the area and by replacing detailed routing information with a single default route.

LSAs Allowed and Blocked in an OSPFv3 Totally Stubby Area

In a totally stubby area, routers continue to receive LSAs necessary to understand **local topology**, but **both external and inter-area routing information is suppressed**.

- **Intra-area LSAs Allowed**
  - **LSA Type 1 (Router LSA)**: Describes intra-area topology
  - **LSA Type 8 (Link LSA)**: Supports IPv6 neighbor communication
  - **LSA Type 9 (Intra-Area Prefix LSA)**: Carries IPv4 and IPv6 prefixes within the area

  These LSAs allow routers in the area to fully understand local connectivity and install routes to networks inside the area.
- Default Route Injection Allowed
  - **Single Default Route injected by the ABR**
  - Advertised as a Type 3 Inter-Area Prefix LSA

  This default route replaces all inter-area and external routing information and provides a path out of the area.
- **LSAs Blocked in an OSPFv3 Totally Stubby Area**
  - **LSA Type 3 (Inter-Area Prefix LSAs)**: Suppressed (except the default route)
  - **LSA Type 4 (Inter-Area Router LSAs)**: Blocked
  - **LSA Type 5 (AS-External LSAs)**: Blocked

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Because neither inter-area prefixes nor external routes are allowed, there is no need to advertise how to reach others' areas or the ASBR.

What We Expect to See in a Totally Stubby Area

When Area 2 is configured as a totally stubby area, internal routers such as **D2** should have:

- Routes to OSPF networks **within Area 2**
- **A single default route injected by the ABR**
- No visibility into individual inter-area or external prefixes

This default route provides connectivity to all destinations outside the area without carrying detailed routing information.

What We Do Not Expect to See in a Totally Stubby Area

Because additional LSAs are suppressed, routers inside Area 2 should **not** have:

- Inter-area routes from other OSPF areas
- The default route advertised by the ASBR
- External routes, including redistributed IPv6 static routes

Summary

A totally stubby area further reduces routing information compared to a standard stub area by allowing only intra-area LSAs and a single default route from the ABR. By blocking both inter-area and external LSAs, OSPFv3 minimizes LSDB size and routing table entries while maintaining full reachability through the ABR.

This sets the stage for comparing totally stubby areas with NSSA and totally NSSA areas in later sections of the lab.

Step 1: Configure Area 2 as a Totally Stubby Area on ABR router R3

The ABR is configured as part of a totally stubby area by applying the **area** <*area-id*> **stub no-summary** command under the appropriate **OSPFv3 address family**. This time, adjacencies do not reset.

```
R3(config)# router ospfv3 123
R3(config-router)# address-family ipv4 unicast
R3(config-router-af)# area 2 stub no-summary
R3(config-router-af)# exit-address-family
R3(config-router)# address-family ipv6 unicast
R3(config-router-af)# area 2 stub no-summary
R3(config-router-af)# exit-address-family
```

Step 2: Configure Area 2 as a Totally Stubby Area on Internal router D2

Internal routers in a totally stubby area are configured the same as an internal router in a stub area. Therefore, there is no change for D2.

```
D2(config)# router ospfv3 123
D2(config-router)# address-family ipv4 unicast
D2(config-router-af)# area 2 stub
D2(config-router-af)# exit-address-family
D2(config-router)# address-family ipv6 unicast
D2(config-router-af)# area 2 stub
```

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```
D2(config-router-af)# exit-address-family
```

Step 3: Verify routing tables on D2

After Area 2 is configured as a **totally stubby area**, the routing tables on D2 show a further reduction in routing information compared to a standard stub area. As expected, D2 no longer receives individual **inter-area routes** or any **external routes**. Instead, D2 installs **only a single default route** learned from the ABR (R3), which replaces all inter-area and external routing information.

This behavior reflects the LSA filtering performed by the ABR in a totally stubby area. D2 continues to receive **intra-area LSAs**, including Router LSAs (Type 1), Link LSAs (Type 8), and Intra-Area Prefix LSAs (Type 9), allowing it to fully understand the topology and prefixes within Area 2. However, **Type 3 summary LSAs for specific inter-area prefixes, Type 4 LSAs, and Type 5 external LSAs are all suppressed**, leaving only a **default summary route (Type 3)** injected by the ABR.

As shown in the output, both the IPv4 and IPv6 routing tables on D2 now contain **only a default route** pointing to the ABR, with no visibility into individual networks in other areas or any externally redistributed routes.

Note: Intra-area OSPFv3 routes **(O)** would be included, however there are only directly connected networks for D2 in Area 2.

```
D2# show ip route ospfv3 | begin Gateway
Gateway of last resort is 10.10.4.1 to network 0.0.0.0

O*IA  0.0.0.0/0 [110/2] via 10.10.4.1, 00:00:45, GigabitEthernet1/0/11
D2#
D2# show ipv6 route ospf | begin ::/0
OI  ::/0 [110/2]
     via FE80::3:1, GigabitEthernet1/0/11
```

Although the routing information available to D2 has been significantly reduced, again, **overall network reachability has not changed**. The ABR, R3, still receives full inter-area and external routing information, including the default route advertised by the ASBR, R2. Traffic from D2 destined for networks outside Area 2 is forwarded to the ABR using the injected default route, and the ABR then forwards the traffic based on its complete routing table.

This demonstrates that configuring a totally stubby area **simplifies the control plane** by minimizing LSAs and routing table entries even more, while the **data plane continues to forward traffic normally**, preserving end-to-end connectivity.

Part 6: OSPFv3 NSSA (Not-So-Stubby Areas)

Unlike a stub area, an NSSA **permits an internal router to function as an ASBR**, while still preventing Type 5 external LSAs from entering the area. This makes NSSA areas useful at the edge of an OSPF domain, where redistribution is required but full external flooding is not desirable.

LSAs Allowed and Blocked in an OSPFv3 NSSA

An NSSA allows most of the same LSAs as a stub area, with one key addition.

- **Intra-area LSAs Allowed**
  - **LSA Type 1 (Router LSA)**: Describes intra-area topology
  - **LSA Type 8 (Link LSA)**: Supports IPv6 neighbor communication
  - **LSA Type 9 (Intra-Area Prefix LSA)**: Carries IPv4 and IPv6 prefixes within the area

  These LSAs allow routers to fully understand local topology and prefixes.

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- **Inter-area LSAs Allowed**
  - **LSA Type 3 (Inter-Area Prefix LSA)**: Advertised by the ABR for networks in other OSPF areas
  - **Default Summary LSA (Type 3)**: Injected by the ABR to provide a default route
  - **NSSA-Specific External LSAs (Allowed)**
  - **LSA Type 7 (NSSA External LSA)**: Generated by an ASBR inside the NSSA

  Type 7 LSAs allow external routes to exist *inside* the NSSA without flooding Type 5 LSAs.
- **LSAs Blocked in an OSPFv3 NSSA**
  - **LSA Type 5 (AS-External LSA)**: Blocked inside the NSSA
  - **LSA Type 4 (Inter-Area Router LSA)**: Not required inside the NSSA

  Instead of Type 5 LSAs, external routes injected within the NSSA are carried as **Type 7 LSAs**.

What We Expect to See in an NSSA

When Area 1 is configured as an NSSA and D1 is configured as an NSSA ASBR, routers inside the area should have:

- Routes to OSPF networks within Area 1
- Routes to OSPF networks in other areas (via Type 3 LSAs)
- A default route injected by the ABR
- External routes injected *inside* the NSSA, appearing as **Type 7 LSAs**

What We Do Not Expect to See in an NSSA

- External routes advertised as **Type 5 LSAs** inside Area 1
- Type 5 LSAs flooding directly into the NSSA from other areas

What We Expect to See in other OSPFv3 Areas

In the **other normal OSPF areas (including Area 0)**, you should expect the following as a result of **Area 2 being an NSSA with D2 redistributing a static route**:

- The **ABR translates the NSSA Type 7 LSA into a Type 5 AS-External LSA** when advertising the route into normal areas.
- Routers in normal areas will therefore see the redistributed route as a **normal external OSPF route (OE1 or OE2)**, just as if it had been injected by an ASBR in a normal area.
- These routers will also have **reachability to the NSSA ASBR** via the ABR (using internal OSPF topology), but they do **not** see any Type 7 LSAs directly.

What does this mean conceptually?

- **Inside Area 1 (NSSA)**: External routes appear as **Type 7 (ON1/ON2)**.
- **Outside Area 1 (normal areas)**: The same routes appear as **Type 5 (OE1/OE2)** after translation by the ABR.

In this part, you will configure **D1 as an NSSA ASBR** within Area 1. Area 1 represents a branch site connected to the backbone through R1 over a bandwidth-constrained link. Router D1 operates as an NSSA ASBR, simulating a local Internet connection by redistributing an external IPv6 route into OSPF. Area 1 is configured as an NSSA to remain mostly stub-like, blocking external LSAs from Area 0 while still allowing D1 to inject local external routes as **Type 7 LSAs** that R1 can translate and advertise to the rest of the OSPF domain.

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Step 1: Configure Loopback 1 interface on D1 in Area 1 to serve as an NSSA ASBR.

Configure a loopback interface with IPv6 addresses on **D1** to simulate a connection to an external network.

```
D1(config)# interface Loopback0
D1(config-if)# ipv6 address fe80::d2:3 link-local
D1(config-if)# ipv6 address 2001:db8:feed:211::1/64
D1(config-if)# no shutdown
D1(config-if)# exit
```

Step 2: Configure an IPv6 static route on D1.

Configure an IPv6 static route that points to the loopback interface. This static route simulates an external network reachable through D1 and provides a prefix that can be redistributed into OSPFv3. Without this route, D1 would have no external reachability information to advertise, and it would not function as an NSSA ASBR.

```
D1(config)# ipv6 route 2001:db8:cafe:1::/64 loopback0
```

Step 3: Configure Area 1 as an NSSA on D1.

Internal routers in an NSSA must also be configured with the NSSA area type. Since Area 1 is IPv6 only, we only need to configure the **area 1 nssa** command for the IPv6 address family. Adjacency with R1 will be reset.

```
D1(config)# router ospfv3 123
D1(config-router)# address-family ipv6 unicast
D1(config-router-af)# area 1 nssa
*Jan 16 18:27:31.165: %OSPFv3-5-ADJCHG: Process 123, IPv6, Nbr 1.1.1.1 on
GigabitEthernet1/0/11 from FULL to DOWN, Neighbor Down: Adjacency forced to reset
```

Step 4: Configure D1 as an NSSA ASBR.

To demonstrate NSSA behavior, D1 is configured as an ASBR inside the NSSA by redistributing a static route into OSPFv3. This causes D1 to generate **Type 7 LSAs** instead of Type 5 LSAs.

```
D1(config-router-af)# redistribute static
D1(config-router-af)# exit-address-family
```

Step 5: Configure Area 1 as an NSSA on the ABR R1.

The ABR must be configured to recognize Area 1 as an NSSA so that it can properly handle Type 7 LSAs and translate them when advertising routes into other areas. Since Area 1 is IPv6 only, we only need to configure the **area 1 nssa** command for the IPv6 address family, but we have configured it for both AFs in case we later add IPv4 prefixes to Area 1. Notice that adjacency with D1 is reestablished.

```
R1(config)# router ospfv3 123
R1(config-router)# address-family ipv4 unicast
R1(config-router-af)# area 1 nssa
R1(config-router-af)# exit-address-family
R1(config-router)# address-family ipv6 unicast
R1(config-router-af)# area 1 nssa
R1(config-router-af)# exit-address-family
*Jan 16 18:33:36.513: %OSPFv3-5-ADJCHG: Process 123, IPv6, Nbr 1.1.1.2 on
GigabitEthernet0/0/1 from LOADING to FULL, Loading Done
```

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Step 6: Verify the IPv6 Routing Table on D1

After Area 1 is configured as an NSSA and D1 is acting as an NSSA ASBR, the routing tables on D1 reflect the unique behavior of an NSSA. Notice that D1 is **no longer receiving external prefixes, LSA 5s**, from the ASBR R2, because they are being blocked by the ABR R1.

```
D1# show ipv6 route | begin 2001
C   2001:DB8:ACAD:1001::/64 [0/0]
     via GigabitEthernet1/0/11, directly connected
L   2001:DB8:ACAD:1001::2/128 [0/0]
     via GigabitEthernet1/0/11, receive
C   2001:DB8:ACAD:1002::/64 [0/0]
     via GigabitEthernet1/0/23, directly connected
L   2001:DB8:ACAD:1002::1/128 [0/0]
     via GigabitEthernet1/0/23, receive
OI  2001:DB8:ACAD:2001::/64 [110/4]
     via FE80::1:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:2002::/64 [110/5]
     via FE80::1:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:A001::/64 [110/2]
     via FE80::1:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:A002::/64 [110/3]
     via FE80::1:1, GigabitEthernet1/0/11
S   2001:DB8:CAFE:1::/64 [1/0]
     via Loopback0, directly connected
C   2001:DB8:FEED:211::/64 [0/0]
     via Loopback0, directly connected
L   2001:DB8:FEED:211::1/128 [0/0]
     via Loopback0, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

From D1's routing table, we can observe the following:

- **Connected and local routes** for D1's directly attached IPv6 networks appear as expected.
- **OSPF inter-area routes (OI)** are present for networks in Area 0 and Area 2. These routes are learned via **Type 3 Inter-Area Prefix LSAs** generated by the ABR R12.
- **No external OSPF routes (OE1/OE2)** appear in the routing table. This is because **Type 5 AS-External LSAs are blocked from entering an NSSA**.
- The static IPv6 route configured on D1 (2001:db8:cafe:1::/64) appears as a **local static route**. However, OSPF now originates a **Type 7 LSA** for this prefix within Area 1, allowing other routers in the NSSA to learn the route without violating stub-area behavior.

In short, D1 has full visibility of the **internal OSPF topology and inter-area prefixes**, but is **isolated from external routing information**, which is the defining behavior of an NSSA.

Step 7: Configure ABR R1 as a default route into the NSSA Area 1.

To inject a default route into Area 1 (the NSSA) from the ABR R1, use the default-information-originate option on the NSSA area command under the appropriate OSPFv3 address family.

In an NSSA, the ABR injects a default route using a Type 7 NSSA External LSA, not a Type 3 Summary LSA. This causes the default route to appear as an ON1 or ON2 route on internal NSSA routers, ensuring that all external reachability inside the NSSA is handled consistently using Type 7 LSAs.

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```
R1(config)# router ospfv3 123
R1(config-router)# address-family ipv6 unicast
R1(config-router-af)# area 1 nssa default-information-originate
R1(config-router-af)# exit-address-family
```

Step 8: Verify the default route in the IPv6 Routing Table on D1

After configuring **R1 (the ABR)** to inject a default route into **Area 1 (NSSA)**, D1 now learns an IPv6 default route shown as **ON2 ::/0**. The **ON2** code indicates that this route is an **OSPF NSSA external Type 2 route**, carried using a **Type 7 LSA**.

Even though the default route is injected by the ABR, it is still advertised into the NSSA using **Type 7 LSAs**, not Type 3 Summary LSAs. This is specific to NSSA behavior: default routes injected into an NSSA appear as **NSSA external routes (Type 7)** so they are treated consistently with other external information inside the area.

Because this route is **Type 2**, the metric shown ([110/1]) represents only the **external cost**, and the internal cost to reach the ABR is calculated separately. This mirrors the behavior of **OE2 routes** in normal areas but uses **Type 7 LSAs instead of Type 5**.

```
D1# show ipv6 route ospf | begin ON2
       ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-registrations
       ld - LISP dyn-eid, lA - LISP away, le - LISP extranet-policy
       lp - LISP publications, ls - LISP destinations-summary
ON2 ::/0 [110/1]
     via 2001:DB8:ACAD:1001::1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:2001::/64 [110/4]
     via FE80::1:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:2002::/64 [110/5]
     via FE80::1:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:A001::/64 [110/2]
     via FE80::1:1, GigabitEthernet1/0/11
OI  2001:DB8:ACAD:A002::/64 [110/3]
     via FE80::1:1, GigabitEthernet1/0/11
```

Step 9: Verify R3 has the external route from D1 in its routing table.

The IPv6 routing table on **R3**, which resides in **Area 0**, shows the combined effects of **two different ASBRs** in the OSPF domain: the original ASBR in Area 0 (R2) and the **NSSA ASBR in Area 1 (D1)**.

```
R3# show ipv6 route ospf | begin ::/0
OE2 ::/0 [110/1], tag 123
     via FE80::2:2, GigabitEthernet0/0/0
OI  2001:DB8:ACAD:1001::/64 [110/3]
     via FE80::2:2, GigabitEthernet0/0/0
OI  2001:DB8:ACAD:1002::/64 [110/4]
     via FE80::2:2, GigabitEthernet0/0/0
O   2001:DB8:ACAD:2002::/64 [110/2]
     via FE80::D2:2, GigabitEthernet0/0/1
O   2001:DB8:ACAD:A001::/64 [110/2]
     via FE80::2:2, GigabitEthernet0/0/0
OE2 2001:DB8:CAFE:1::/64 [110/20]
     via FE80::2:2, GigabitEthernet0/0/0
OE2 2001:DB8:FACE:1::/64 [110/20]
```

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```
        via FE80::2:2, GigabitEthernet0/0/0
```

From the output, R3 learns:

- **OE2 ::/0** - This default route is advertised by the **Area 0 ASBR (R2)** as a **Type 5 AS-External LSA**. Because R3 is in Area 0, it is permitted to receive external LSAs directly, and the route appears as an **OE2** entry.
- **OE2 2001:DB8:FACE:1::/64** - This route represents the IPv6 static route redistributed by the **Area 0 ASBR (R2)** and is carried throughout the OSPF domain as a **Type 5 LSA**.
- **OE2 2001:DB8:CAFE:1::/64** - This route originated inside **Area 1 (NSSA)** on D1 as a **Type 7 LSA**. The ABR (R1) translated the Type 7 LSA into a **Type 5 AS-External LSA** when advertising it into Area 0, which is why R3 sees it as an **OE2** route.
- **OI routes (Inter-Area)** - Prefixes from other areas are learned via **Type 3 Inter-Area Prefix LSAs**, explaining the presence of multiple **OI** entries.
- **O routes (Intra-Area)** - Prefixes local to Area 0 are learned via **Type 1 and Type 9 LSAs**, resulting in **O** entries in the routing table.

Summary

An NSSA blocks Type 5 external LSAs like a stub area, reducing routing overhead, but still allows an internal ASBR to inject external routes as Type 7 LSAs. The ABR translates these Type 7 LSAs into Type 5 LSAs when advertising them to other OSPF areas and provides a default route so routers in the NSSA can reach external destinations.

Part 7: OSPFv3 Totally NSSA

In this part, you will configure **Area 1 as an OSPFv3 Totally NSSA** and examine how it further restricts routing information compared to a standard NSSA. A Totally NSSA combines the characteristics of a totally stubby area and an NSSA: it allows local redistribution inside the area while aggressively limiting the routing information learned from outside the area.

This area type is useful when an edge area must inject local external routes into OSPF but does not need visibility into inter-area or external routes from elsewhere in the OSPF domain. Like other stub-based areas, a Totally NSSA relies on a default route injected by the ABR to maintain full reachability.

Difference between NSSA and Totally NSSA

A **Totally NSSA** behaves the same as an NSSA **with one important difference**: it further reduces routing information by **suppressing inter-area routes – no Type 3 LSAs**.

In a standard **NSSA**, routers receive:

- Intra-area routes
- Inter-area routes (**Type 3 LSAs**)
- An **optional default route** injected by the ABR **(Type 7 LSA**)
- Local external routes as **Type 7 LSAs**

In a **Totally NSSA**, routers receive:

- Intra-area routes only
- An **optional default route** injected by the ABR **(Type 7 LSA**)
- Local external routes as **Type 7 LSAs**

All **inter-area routes (Type 3 LSAs) are suppressed**, and routers rely entirely on the default route to reach destinations outside the area.

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Step 1: Modify R1 to change Area 1 from NSSA to Totally NSSA and keep default route injection.

Under each address family, add **no-summary** and keep **default-information-originate**. You can either replace it directly (IOS will often accept the new form), but explicitly removing the old line first is better:

```
R1(config)# router ospfv3 123
R1(config-router)# address-family ipv6 unicast
R1(config-router-af)# no area 1 nssa default-information-originate
R1(config-router-af)# area 1 nssa no-summary default-information-originate
R1(config-router-af)# exit-address-family
```

Notes:

- The command **no area 1 nssa default-information-originate** is not required but recommended.
- The command **area 1 nssa no-summary default-information-originate** would be repeated under address-family ipv4 unicast if you are also doing IPv4 AF.

Step 2: There is no change to D1.

D1 should still be configured as **NSSA** (same as before):

```
D1# show run | section router ospfv3 123
router ospfv3 123
 !
 address-family ipv6 unicast
  passive-interface GigabitEthernet1/0/23
  redistribute static
  router-id 1.1.1.2
  area 1 nssa
 exit-address-family
```

Step 3: Verify the IPv6 Routing Table on D1

After Area 1 is configured as a **Totally NSSA**, the IPv6 routing table on **D1** shows a significantly reduced set of routes, while still maintaining full reachability.

```
D1# show ipv6 route | begin ::/0
OI  ::/0 [110/2]
     via FE80::1:1, GigabitEthernet1/0/11
C   2001:DB8:ACAD:1001::/64 [0/0]
     via GigabitEthernet1/0/11, directly connected
L   2001:DB8:ACAD:1001::2/128 [0/0]
     via GigabitEthernet1/0/11, receive
S   2001:DB8:CAFE:1::/64 [1/0]
     via Loopback0, directly connected
C   2001:DB8:FEED:211::/64 [0/0]
     via Loopback0, directly connected
L   2001:DB8:FEED:211::1/128 [0/0]
     via Loopback0, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

From the routing table, we can observe the following:

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- **Intra-area routes** - D1 retains its directly connected and local IPv6 networks. These routes are learned and maintained through **intra-area LSAs (Type 1, Type 8, and Type 9)**, allowing D1 to fully understand topology and prefixes within Area 1.
- **Default route only for external reachability** - D1 now has a **single IPv6 default route (::/0) (Type 7 LSA)** pointing to the **ABR (R1)**. This default route replaces all inter-area and external prefixes and provides reachability to all destinations outside Area 1. Because inter-area routes are suppressed in a Totally NSSA, the default route is the only path D1 uses for traffic leaving the area.
- **No inter-area or external-specific routes** - Individual inter-area prefixes **(Type 3 LSAs)** and external routes (Type 5 LSAs) do not appear in the routing table. This is a defining characteristic of a Totally NSSA and explains why the routing table is much smaller than in a normal area or standard NSSA.
- **Local static route remains visible** - The locally configured static route (2001:db8:cafe:1::/64) is still present because it is directly configured on D1. When redistributed, this route is advertised to the rest of the OSPF domain as an NSSA external route **(Type 7)**, even though it does not appear as an OSPF-learned route on D1 itself.

Note: When D1 and R1 are configured to make Area 1 a Totally NSSA, this change affects only the routing information inside Area 1 and has no effect on the routing tables of routers in Area 0 or Area 2.

Part 8: Reachability for All Areas

In this final section, you will tie together everything learned in this lab by examining how **OSPFv3 area types affect routing information without affecting reachability**. Rather than memorizing what each area type "allows" or "blocks," understanding **which LSAs are generated, filtered, or replaced** explains *why* each area behaves the way it does. By focusing on LSAs and their role in the control plane, you can reason through any OSPF design and clearly see how OSPF preserves full data-plane connectivity even as routing visibility is intentionally reduced to improve scalability.

Summary of LSAs: Normal, Stub and Totally Stubby Areas

The following table summarizes how different OSPFv3 area types control routing information by permitting or blocking specific Link-State Advertisements (LSAs). By comparing normal, stub, and totally stubby areas, you can see how OSPF progressively reduces the amount of routing information shared within an area and replaces it with a default route injected by the ABR. This comparison reinforces how OSPF area types influence the control plane while preserving full data-plane reachability.

This is a summary of LSAs received by an OSPFv3 internal router in each of these areas.

LSA Type	Normal Area	Stub Area	Totally Stubby Area	NSSA	Totally NSSA
Type 1 (Router LSA)	✓ Received	✓ Received	✓ Received	✓ Received	✓ Received
Type 8 (Link LSA)	✓ Received	✓ Received	✓ Received	✓ Received	✓ Received
Type 9 (Intra-Area Prefix LSA)	✓ Received	✓ Received	✓ Received	✓ Received	✓ Received
Type 3 (Inter-Area Prefix LSA)	✓ Received (all summaries)	✓ Received (all summaries)	❌ Suppressed	✓ Received (all summaries)	❌ Suppressed
Type 3 Default Route from ABR	❌ Not needed	✓ Injected by ABR	✓ Injected by ABR	❌ Replaced by Type 7 default	❌ Replaced by Type 7 default
Type 4 (Inter-Area Router LSA)	✓ Received	❌ Blocked	❌ Blocked	❌ Blocked	❌ Blocked

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Type 5 (AS-External LSA)	✓ Received	❌ Blocked	❌ Blocked	❌ Blocked	❌ Blocked
Type 7 (NSSA External LSA)	❌ Not used	❌ Not used	❌ Not used	✓ Received (local externals + default)	✓ Received (local externals + default)

Stub and totally stubby areas receive an ABR-injected default route automatically, while NSSA and Totally NSSA areas require explicit ABR configuration to inject a default route.

Area Type	Default Route Injected Automatically?	Default LSA Type
Stub	✅ Yes	Type 3
Totally Stubby	✅ Yes	Type 3
NSSA	❌ No (must configure ABR)	Type 7
Totally NSSA	❌ No (must configure ABR)	Type 7

Reachability in a Normal OSPF Area

Routers in a **normal OSPF area** have **full reachability to all networks** because they also have **full routing visibility**. A normal area receives intra-area, inter-area, and external routes, including routes injected by ASBRs and routes translated from NSSA Type 7 LSAs into Type 5 LSAs by an ABR. Because no LSAs are filtered in a normal area, routers can forward traffic using the most specific available routes.

Normal areas participate fully in OSPF routing and provide complete visibility into the OSPF domain, allowing traffic to be forwarded accurately and efficiently without relying on default routes.

Reachability in a Stub OSPF Area

Routers in a **stub OSPF area** also maintain **full reachability to all networks**, even though they receive less routing information than routers in a normal area. In a stub area, external routes are blocked, and routers do not receive Type 5 AS-External LSAs. Instead, the ABR automatically injects a **default route using a Type 3 Summary LSA**, which provides reachability to destinations outside the area.

Inter-area routes are still permitted in a stub area, allowing routers to learn specific prefixes from other OSPF areas while relying on the default route for external destinations. This design reduces external routing information while preserving complete connectivity through the ABR.

Reachability in a Totally Stubby OSPF Area

Routers in a **totally stubby OSPF area** also maintain **full reachability to all networks**, even though they have the most limited routing visibility. These routers receive only **intra-area routes** and a **single default route injected by the ABR**, with all inter-area and external routes suppressed. This significantly reduces routing table size and LSDB complexity.

When traffic is destined for networks outside the area, it is forwarded to the ABR using the default route. The ABR—having full routing knowledge—then forwards the traffic appropriately. This demonstrates that totally stubby areas simplify the **control plane** while leaving the **data plane** fully functional.

Reachability in an NSSA OSPF Area

Routers in an **NSSA OSPF area** also have **full reachability to all networks**, but they achieve it with less routing information than a normal area. External routes originating outside the NSSA are blocked, so routers do not see detailed external prefixes. Instead, the ABR can be configured to inject a **default route using a Type 7 LSA**, allowing traffic to reach destinations outside the area without carrying full external routing details.

At the same time, an NSSA supports **local redistribution**. Routes injected by an NSSA ASBR are advertised inside the area as **Type 7 LSAs** and translated by the ABR into **Type 5 LSAs** for the rest of the OSPF domain. This preserves full end-to-end reachability while limiting external LSA flooding.

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Reachability in a Totally NSSA OSPF Area

Routers in a **totally NSSA OSPF area** also maintain **full reachability to all networks**, while carrying even less routing information than a standard NSSA. In this area type, **inter-area routes are suppressed**, and routers rely entirely on a **single default route injected by the ABR (optional)** for destinations outside the area. This further reduces routing table size and LSDB complexity.

Despite this reduced visibility, local redistribution is still supported. External routes injected by an NSSA ASBR are carried within the area as **Type 7 LSAs** and translated by the ABR into **Type 5 LSAs** for other areas. This shows that a totally NSSA provides the most restrictive control-plane behavior while still preserving full data-plane connectivity.

Summary

Across **all OSPF area types**, reachability to all networks is preserved. What changes is **how that reachability is achieved**:

- **Normal areas** use full routing information, including intra-area, inter-area, and external routes.
- **Stub areas** rely on inter-area routes and an **ABR-injected default route** to reach external networks.
- **Totally stubby areas** depend entirely on a **single default route injected by the ABR** for all destinations outside the area.
- **NSSAs** rely on **Type 7 LSAs for locally redistributed routes** and an **ABR-injected default route** for other external destinations.
- **Totally NSSAs** use **Type 7 LSAs for local redistribution** and depend solely on a **single default route injected by the ABR** for all inter-area and external reachability.

OSPF area design controls **routing visibility**, not **end-to-end connectivity**, allowing large networks to scale efficiently without sacrificing reachability.

Summary of reachability:

Area Type	Routing Information Visibility	How Full Reachability Is Achieved	Summary
Normal Area	Full visibility: intra-area, inter-area, and external routes (LSA Types 1, 3, 4, 5, 8, 9)	Uses **specific routes**, including external routes learned via **Type 5 LSAs** from ASBRs and translated NSSA routes	Normal areas have complete routing information and forward traffic using the most specific routes available.
Stub Area	Reduced visibility: intra-area and inter-area routes only (LSA Types 1, 3, 8, 9); no Type 5 LSAs	Uses a **default route injected by the ABR (Type 3)** for all external destinations	Stub areas reduce external routing information while preserving full reachability via the ABR.
Totally Stubby Area	Minimal visibility: intra-area routes only (LSA Types 1, 8, 9) plus a single default route	Uses a **single default route injected by the ABR (Type 3)** to reach all inter-area and external networks	Totally stubby areas minimize routing tables and LSDB size while relying entirely on the ABR.
NSSA	Limited visibility: intra-area, inter-area, and NSSA external routes (LSA Types 1, 3, 7, 8, 9); no Type 5 LSAs	Uses an **ABR-injected default route (Type 7)** for external reachability and **Type 7 LSAs** for locally redistributed routes	NSSA areas allow local redistribution while limiting external LSA flooding.
Totally NSSA	Minimal visibility: intra-area routes only (LSA Types 1, 8, 9)	Uses a **single default route injected by the ABR (Type 7)** and **Type 7 LSAs** for local redistribution	Totally NSSA areas provide the most restrictive control plane

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	plus NSSA externals (Type 7) and a default route		while still maintaining full reachability.

Final Word

In real networks, the choice of OSPF area type and how external routes are advertised is driven by overall network design. Factors such as the presence of **multiple ABRs for an area** or **multiple ASBRs** injecting external routes can influence these decisions. In these scenarios, using **External Type 1 routes** may be preferred, because they include the internal OSPF cost to reach the ASBR, allowing routers to select the true lowest end-to-end path rather than relying only on the external metric. Understanding how LSAs are filtered and how external metrics are calculated helps network engineers design scalable OSPF deployments that make predictable and optimal forwarding decisions.

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